

CIS530 HW9: Submission Template

Text Simplification as Machine Translation

Problem 1.2: General Changes for Text Simplification (5 points)

Based on my experience manually simplifying complex sentences, here are five general changes that can be made to transform complex sentences into simpler ones:

1. **Remove dependent clauses:** Cut out subordinate clauses (e.g., "although," "because", "while") and keep only the main idea.
 2. **Use simpler vocabulary:** Replace advanced or technical words with more common, everyday terms.
 3. **Split into two or more short sentences:** Break up long, multi-clause sentences into shorter, stand-alone ones.
 4. **Eliminate unnecessary modifiers or qualifiers:** Remove extra details, adjectives, or adverbs that don't change the core meaning.
 5. **Use active voice instead of passive:** Change passive constructions (e.g., "was designed by") to active ones (e.g., "designed").
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Problem 6.1: Evaluation and Analysis (5 points)

Overall Performance Comparison

Based on human evaluation of 10 randomly selected complex sentences, the fine-tuned model consistently produced the best simplifications. Its outputs were the most grade-appropriate, concise, and easy to understand. The few-shot approach also performed reasonably well, benefiting from the in-context examples. In contrast, the zero-shot approach often failed to simplify effectively, producing outputs that were either too verbose or too advanced for the target reading level. Overall, fine-tuning was the most reliable and consistent method, especially for educational applications.

Zero-shot Performance

The zero-shot model had limited success. While it generated fluent English responses, they were often not simplified enough for a grade 4 reader. Some sentences retained difficult

vocabulary or complex sentence structures. It also lacked consistency — some outputs were close to readable for younger students, while others barely simplified the input at all. A key weakness was that without explicit examples or training, the model struggled to understand what “simplify for a student” truly meant.

Few-shot Performance

The few-shot model performed significantly better than zero-shot. By including a couple of good simplification examples, the model better understood both the tone and structure expected in the output. The few-shot completions used simpler vocabulary and shorter sentences more consistently. However, it still struggled with more complex or unfamiliar sentence types, especially when the examples weren't directly analogous. Still, the few-shot approach showed that well-chosen demonstrations can noticeably improve output quality, even without model retraining.

Fine-tuning Performance

The fine-tuned model produced the most accurate and appropriate simplifications. Its responses consistently matched the target grade level, avoiding overly technical vocabulary and breaking down complex ideas clearly. Compared to prompting methods, fine-tuning led to much more consistent and robust performance, especially across a variety of sentence types. The model clearly learned the simplification style and format from the training data, generalizing well to new examples that were not part of its training set.

Expected vs. Unexpected Results

I expected the fine-tuned model to outperform the others — and it did. However, I was pleasantly surprised by the improvement from zero-shot to few-shot. Just two examples had a meaningful effect on output quality, showing how powerful in-context learning can be. One unexpected result was how fluency in zero-shot output often masked the lack of simplification, which highlighted the need for careful human evaluation. Overall, the results aligned with expectations but also emphasized the impact of good prompt design.

Key Insights and Learnings

- This evaluation provided valuable insights into the trade-offs between prompt engineering and fine-tuning:
- Fine-tuning is ideal when consistency, control, and task-specific formatting are important.
- Few-shot prompting is an effective and flexible alternative when fine-tuning isn't available.
- Zero-shot prompting is convenient but unreliable for tasks that require specific constraints, like reading-level simplification.

- Human evaluation is essential — models can sound fluent while failing the task criteria.
- Text simplification is not just about shortening sentences — it's about rephrasing in a cognitively accessible way, which requires deeper understanding from the model.

Limitations and Future Improvements

A limitation of this evaluation is that it relied on human subjective judgment for selecting the “best” output. A more robust evaluation could include multiple annotators, readability metrics, or even user testing with actual students. Additionally, the fine-tuned model was only trained on simplified titles, not full articles — training on full texts may yield better generalization and depth of simplification.

Summary

In conclusion, fine-tuning clearly provided the best results for text simplification to a target reading level, followed by few-shot prompting, and then zero-shot. Each approach has its trade-offs, but for consistent and effective educational simplifications, fine-tuning is the most robust option. This exercise also reinforced the importance of task-specific data, strong evaluation design, and understanding the nuances of the simplification task.